The Geometric Vacuum: Quantum Structure from Compact Manifold Topology*

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The non-relativistic quantum mechanics of Coulomb systems can be reformulated as spectral geometry on compact manifolds. The angular momentum quantum numbers (l, m) and their degeneracies $2l + 1$ emerge from a geometric packing construction on isotropic shells, independent of the Schrödinger equation. The hydrogen energy spectrum, the $1/r$ Coulomb potential, and the n^2 degeneracy all follow from the Laplace–Beltrami operator on the unit three-sphere S^3, with physical scales entering exclusively through stereographic projection—as demonstrated by Fock in 1935 and verified here through 18 independent symbolic proofs. For N-electron atoms, the framework extends to S^{3N-1}, where the permutation group S_N acting on the hyperspherical angular basis produces chemical periodicity as a representation-theoretic consequence. A new sparsity mechanism from angular momentum recoupling—$6j$ coefficients producing additional zeros beyond Gaunt selection rules—reduces electron repulsion integral density by 18–26%. The exchange constant taxonomy classifies transcendental content (where π and other transcendentals enter the otherwise rational graph spectrum) as instances of the Weyl–Selberg spectral-geometric conversion. The Hopf fibration $S^1 \to S^3 \to S^2$ of the electron's proven state manifold yields the fine structure constant $\alpha^{-1} = 137.035987$ from spectral invariants of the bundle components, with zero free parameters and p-value 5.2×10^{-9}; the combination rule remains conjectural. These results are mathematically exact for single-center systems and suggest a concrete research program: high-precision atomic spectra on S^{3N-1}, compared with experimental fine structure data, can test the α conjecture computationally.

I. INTRODUCTION

The Schrödinger equation for hydrogen presupposes continuous flat space, a radial Coulomb potential $V(r) = -Z/r$, and quantized energy levels $E_n = -Z^2/(2n^2)$ as boundary-condition eigenvalues. Fock showed in 1935 [?] that these are not independent assumptions: the hydrogen problem on the unit three-sphere S^3 is a free particle, with the $1/r$ potential emerging from stereographic projection and the Rydberg spectrum from the energy-shell constraint $p_0^2 = -2E$. Bargmann [?] identified the underlying SO(4) symmetry, and Barut and Kleinert [?] extended the dynamical algebra to SO(4,2).

These foundational results are well known. What is new in the present work is their extension into a complete computational framework:

1. A geometric packing construction derives the angular quantum numbers (l, m) and their degeneracies $2l + 1$ from two axioms (binary distinguishability and geometric indifference), independent of operators or dynamics (Paper 0 [?]).

2. The discrete graph Laplacian on the resulting lattice converges to the Laplace–Beltrami operator on the unit S^3, verified through 18 independent symbolic proofs (Paper 7 [?]).

3. The framework extends to N-electron atoms on S^{3N-1}, where the permutation group S_N produces chemical periodicity as representation theory (Papers 13, 16 [? ?]).

4. A new sparsity mechanism from angular momentum recoupling—$6j$ coefficients producing additional zeros beyond Gaunt selection rules—yields 18–26% lower electron repulsion integral (ERI) density in coupled bases.

5. The Hopf fibration $S^1 \to S^3 \to S^2$ suggests a topological origin for the electromagnetic coupling constant α (Paper 2 [?], conjectural).

Scope. The reformulation is mathematically exact for single-center Coulomb systems (atoms). For molecules, the multi-center N-body problem does not separate in any known coordinate system; the GeoVac composed architecture provides practical molecular Hamiltonians with characterized accuracy (∼0.2–5% energy error at fixed geometry), but the single-center mathematical structure is the focus of this paper.

II. KINEMATICS: QUANTUM NUMBERS FROM GEOMETRY

The angular structure of quantum mechanics—the quantum numbers (l, m), the degeneracy $2l+1$ per shell, and the cumulative count n^2—can be derived from two axioms without reference to the Schrödinger equation, angular momentum operators, or any specific physical system.

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A. Two axioms

1. **Binary distinguishability.** A minimum of two distinguishable states is required to establish a nonzero fundamental scale d_0. A single point cannot define a distance; two points create the first bit of spatial information.

2. **Geometric indifference.** In the absence of external constraints or privileged directions, the distribution of states must maximize entropy. By the principle of indifference, this necessitates isotropy: states populate isotropic shells (circles) rather than anisotropic tilings (squares, hexagons), which would imply privileged directions not present in the axioms.

B. Packing construction

The two-point initialization fixes the fundamental area per state $\sigma_0 = \pi d_0^2/2$. States are distributed on linearly spaced shells with radius $r_k = k \cdot d_0$, $k = 1, 2, 3, \ldots$ The annular area of shell k is

$$A_k = \pi d_0^2 \big[k^2 - (k-1)^2\big] = \pi d_0^2 (2k-1), \tag{1}$$

and the angular capacity per shell is

$$N_k^{\text{angular}} = \frac{A_k}{\sigma_0} = 2(2k-1). \tag{2}$$

The factor $(2k-1)$ is the dimension of the spin-$(k-1)$ irreducible representation of SO(3). With the identification $l = k-1$, each shell carries exactly $2l+1$ angular states—the spherical harmonic basis Y_{lm} with $m = -l, \ldots, +l$. The global factor of 2 arises from orientability of the compactified packing manifold on S^2.

The cumulative state count through shell n is

$$\sum_{k=1}^{n} (2k-1) = n^2, \tag{3}$$

reproducing the n^2 orbital count (or $2n^2$ including the spin factor).

C. What the construction establishes

The packing construction derives the *kinematic* structure of quantum mechanics—the state-counting aspect of angular momentum—from pure geometry. It produces the quantum number labeling (l, m), the per-shell degeneracy $2l+1$, and the cumulative count n^2. It does not produce energy eigenvalues, wavefunctions, or dynamics; those require the *dynamic* structure provided by S^3 (Sec. **??**).

III. DYNAMICS: THE SCHRÖDINGER EQUATION FROM S^3

A. The paraboloid lattice

The quantum numbers (n, l, m) from the packing construction label nodes of a discrete graph—the paraboloid lattice. Edges are generated by the ladder operators $L_\pm$ (angular momentum raising/lowering) and $T_\pm$ (radial transitions), producing a sparse adjacency structure. The graph Hamiltonian is

$$H = \kappa \cdot \mathcal{L}, \tag{4}$$

where $\mathcal{L} = D - A$ is the graph Laplacian (D = degree matrix, A = adjacency matrix) and $\kappa = -1/16$ is the universal kinetic scale factor.

B. Continuum limit: the unit S^3

In the continuum limit ($n_{\max} \to \infty$), the graph Laplacian converges to the Laplace–Beltrami operator Δ_{S^3} on the unit three-sphere. The eigenfunctions are Gegenbauer polynomials $C^1_{n-1}(\cos\chi)$ in the $l = 0$ sector, where χ is the hyperspherical angle, and the eigenvalues are pure dimensionless integers:

$$\Delta_{S^3}\, C^1_{n-1}(\cos\chi) = -(n^2-1)\, C^1_{n-1}(\cos\chi). \tag{5}$$

These eigenvalues $\lambda_n = -(n^2-1)$ are topological invariants of the unit sphere, carrying no physical dimensions.

C. Stereographic projection: Fock's map

Fock's stereographic projection [**?**] maps the unit S^3 to flat momentum space $\mathbb{R}^3$. A point $\mathbf{n} = (n_1, n_2, n_3, n_4) \in S^3$ maps to momentum $\mathbf{p} = (p_1, p_2, p_3) \in \mathbb{R}^3$ via

$$n_i = \frac{2p_0\, p_i}{p^2 + p_0^2}, \quad n_4 = \frac{p^2 - p_0^2}{p^2 + p_0^2}, \tag{6}$$

with conformal factor

$$\Omega(\mathbf{p}) = \frac{2p_0}{p^2 + p_0^2}. \tag{7}$$

The unit sphere constraint $\sum_{i=1}^4 n_i^2 = 1$ holds for *all* $p_0 > 0$, confirming scale invariance: the projection always produces the unit S^3 regardless of the choice of p_0.

D. Emergence of the Coulomb potential

The chordal distance identity

$$|\mathbf{n}(\mathbf{p}) - \mathbf{n}(\mathbf{q})|^2 = \Omega(\mathbf{p})\, \Omega(\mathbf{q})\, |\mathbf{p} - \mathbf{q}|^2 \tag{8}$$

relates the Euclidean distance on S^3 (left side) to the flat-space distance (right side), weighted by conformal factors. The free-particle Green's function on S^3, which depends on the chordal distance $|\mathbf{n}-\mathbf{n}'|^{-2}$, transforms under stereographic projection into a kernel proportional to $1/|\mathbf{p}-\mathbf{p}'|^2$. Upon Fourier transformation to position space, this becomes the $1/r$ Coulomb potential.

The Coulomb potential is therefore not an input force law—it is the coordinate distortion introduced by stereographic projection.

E. Energy-shell constraint and the Rydberg spectrum

The parameter p_0 is fixed by the energy-shell constraint

$$p_0^2 = -2E, \tag{9}$$

which converts the dimensionless integer eigenvalues into physical energies. For the n-th level, $p_0 = 1/n$ (in atomic units), and

$$E_n = -\frac{p_0^2}{2} = -\frac{1}{2n^2}. \tag{10}$$

The complete chain is:

$$\text{Graph} \xrightarrow{\text{continuum}} S^3 \xrightarrow{\text{stereo.}} \mathbb{R}^3 \xrightarrow{p_0^2=-2E} E_n = -\frac{1}{2n^2}. \tag{11}$$

F. Symbolic verification

Every algebraic step in this chain has been verified through 18 independent symbolic proofs [?], implemented in the `sympy` computer algebra system and collected in the test suite `test_fock_projection.py` and `test_fock_laplacian.py`. These verify:

- Module I (Tests 1–10): Stereographic projection geometry—unit sphere constraint, conformal factor identities, chordal distance relation, volume Jacobian, inverse projection round-trip.
- Module II (Tests 11–18): Conformal Laplacian and eigenvalues—connection term, zero-mode annihilation ($\Delta_{S^3}(1) = 0$), eigenvalue verification at $n = 2$ and $n = 3$, rational decomposition of conformally weighted Laplacian, eigenvalue-to-energy mapping.

All 18 tests pass with symbolic (exact) agreement, constituting a machine-verified proof of the graph-to-Schrödinger equivalence.

IV. THE MULTI-ELECTRON GENERALIZATION

A. N electrons on S^{3N-1}

For an N-electron atom, the $3N$ spatial coordinates are collected into a hyperradius $R = \sqrt{\sum_i r_i^2}$ and $3N-1$ angular coordinates on S^{3N-1}. The kinetic energy operator separates as

$$T = -\frac{1}{2}\left(\frac{\partial^2}{\partial R^2} + \frac{3N-1}{R}\frac{\partial}{\partial R} - \frac{\Lambda^2}{R^2}\right), \tag{12}$$

where Λ^2 is the grand angular momentum operator (the Laplace–Beltrami operator on S^{3N-1}). Its eigenvalues are the SO($3N$) Casimir invariants:

$$\Lambda^2 Y_\nu = \nu(\nu + 3N - 2)\, Y_\nu, \tag{13}$$

with $\nu = 0, 1, 2, \ldots$ and Y_ν the hyperspherical harmonics.

At hyperradius $R = 0$, the system is a free particle on the unit S^{3N-1}. The total Coulomb interaction enters as a geometric perturbation through the charge function $C(\hat{\Omega})/R$, where C encodes the nuclear attraction and electron-electron repulsion in angular coordinates. In the adiabatic hyperspherical method [? ?], the angular eigenvalue problem is solved parametrically at each R, yielding adiabatic eigenvalues $\mu_\nu(R)$ that define effective radial potentials $V_{\text{eff}}(R) = \mu_\nu(R)/R^2 + (3N-1)(3N-3)/(8R^2)$.

For helium ($N = 2$, S^5), the single-channel adiabatic approximation yields a ground-state energy of -2.9052 Ha, a 0.05% error relative to the exact nonrelativistic value -2.903724 Ha [?].

B. Chemical periodicity from S_N representation theory

The total wavefunction must be antisymmetric under electron permutation. Writing $|\Psi\rangle = |\text{spatial}\rangle \otimes |\text{spin}\rangle$ and applying Schur–Weyl duality, the spatial symmetry is determined by the conjugate of the spin Young diagram.

Theorem 1 (Universal λ_1 [?])**.** *For any N-electron atom with ground-state spin $S < N/2$, the spatial Young diagram has first-column length $\lambda_1 = 2$, independent of Z, S, or electron configuration.*

Proof. The spin diagram has two rows $[n_\uparrow, n_\downarrow]$ with $n_\downarrow \geq 1$ (since $S < N/2$). Its conjugate has $n_\downarrow$ columns of height 2 and $2S$ columns of height 1. The first-column length is 2. □

This fixes the minimum grand angular momentum quantum number at $\nu = N - \lambda_1 = N - 2$, giving a universal free eigenvalue:

$$\boxed{\mu_{\text{free}} = \frac{(N-2)(4N-4)}{2} = 2(N-2)^2.} \tag{14}$$

This "Pauli centrifugal cost" depends only on the electron count N, not on the nuclear charge Z, spin state S, or orbital configuration—it is purely topological.

The periodic table structure follows from the sequence of S_N spatial irreducible representations as shells fill. Five structure types emerge:

Type A: ($N = 1$): Hydrogen. $\nu = 0$, no antisymmetry constraint.

Type B: (Noble gases): Closed-shell, democratic Young diagram $[2^{N/2}]$. Maximal Pauli centrifugal barrier.

Type C: (Alkali metals): One electron in new shell above closed core. Hierarchical diagram $[2^{(N-1)/2}, 1]$.

Type D: (Alkaline earths): Two electrons above closed core. Same diagram shape as Type B but different occupancy pattern.

Type E: (Open-shell): Partially filled p- and d-subshells. Varying spin and diagram shapes.

The periodic law is the group-theoretic statement: *chemical properties of elements are periodic functions of atomic number because the sequence of S_N spatial irreps repeats whenever a new principal shell begins filling.*

C. Angular momentum recoupling sparsity

For N-electron systems, organizing the S^{3N-1} angular basis hierarchically via pairwise (H-set) coupling—where electrons are grouped into pairs with definite intermediate angular momenta $l_{12}, l_{34}, \ldots$—produces a new sparsity mechanism.

When a two-body operator (e.g., the electron-electron repulsion $V_{ee}(1,2)$) acts on a pair, it preserves the angular momenta of spectator pairs. The matrix element of V_{ee} between coupled states involves Wigner $6j$ symbols from recoupling:

$$\begin{Bmatrix} l_1 & l_2 & l_{12} \\ l'_{12} & l'_1 & k \end{Bmatrix}, \tag{15}$$

where k is the multipole order from the Gaunt integral. These $6j$ coefficients vanish for certain combinations of angular momenta, producing *additional zeros* beyond the bare Gaunt selection rules ($|l-l'| \leq k \leq l+l'$, $l+k+l' =$ even).

The resulting ERI density reduction is quantified in Table **??**.

This sparsity mechanism is basis-agnostic: it applies to any encoding with angular momentum pair coupling and follows from the representation theory of SO(3), not from any particular choice of radial basis or physical system.

TABLE I. ERI density (fraction of nonzero electron repulsion integrals) in uncoupled vs. H-set coupled bases for four-electron systems.

$l_{\max}$	Uncoupled	H-set	Reduction
1	12.5%	9.2%	26%
2	5.1%	4.2%	18%

V. WHERE CONTINUOUS PHYSICS ENTERS

A. The π-free graph

The graph Laplacian on the paraboloid lattice is π-free: all eigenvalues, degeneracies, and coupling coefficients are integers or rationals. Transcendental numbers—π, exponential integrals, spectral zeta values—enter exclusively when projecting the discrete graph onto continuous manifolds for comparison with experiment or computational convenience.

This observation motivates a classification of exactly *where* continuous physics enters the otherwise discrete framework.

B. Four types of exchange constant

Each projection from the discrete algebraic structure to a continuous coordinate system introduces a characteristic transcendental constant, classified by what determines it [**?**]:

Intrinsic:: Determined by dimension alone. Example: π for $S^1 \to \mathbb{R}$.

Calibration:: Determined by graph connectivity. Example: the kinetic scale factor $\kappa = -1/16$, which maps the graph eigenvalue $\lambda_{\max} = 8$ to the hydrogen ground state $E_H = -1/2$ Ha.

Embedding:: Seeded by coordinate singularities. Example: the Stieltjes seed $e^a E_1(a)$ that enters when embedding the Laguerre spectral basis into prolate spheroidal coordinates (Paper 11), and the $1/r_{12}$ cusp—an embedding exchange constant that cannot be absorbed into the S^5 graph (Paper 15 [**?**]).

Flow:: Parameter-dependent algebraic functions with transcendental coefficients. Example: the adiabatic eigenvalue $\mu(R)$, which satisfies a polynomial equation $P(R, \mu) = 0$ at each hyperradius but whose R-dependence is transcendental (Paper 13).

These form a dependency hierarchy: intrinsic $\to$ calibration $\to$ embedding $\to$ flow, with each level incorporating the transcendental content of the previous levels.

C. Weyl–Selberg identification

The exchange constants are instances of a known mathematical structure. Weyl's law relates the eigenvalue counting function $N(\Lambda)$ of a compact Riemannian manifold $\mathcal{M}$ to its volume:

$$N(\Lambda) \sim \frac{\mathrm{Vol}(\mathcal{M})}{(2\pi)^d}\,\Lambda^{d/2}, \tag{16}$$

providing the precise conversion between discrete eigenvalue sums and continuous manifold volumes. The Selberg trace formula generalizes this to an exact identity between spectral data and geometric data (lengths of closed geodesics) on hyperbolic manifolds.

The GeoVac exchange constants—κ, $e^a E_1(a)$, $\mu(R)$—are specific instances of this spectral-geometric conversion for the conformal and hyperspherical hierarchies. The exchange constant taxonomy tells us exactly where continuous physics enters the discrete framework, and why it takes the specific transcendental form it does: the graph is π-free; π enters when you project it.

D. Observable classification

The exchange constant hierarchy induces a classification of physical observables by their transcendental content [**?**]:

Class S (Spectral):: Energy eigenvalues, degeneracies, selection rules. Require graph eigenvalues and κ only. *No transcendental content* beyond the calibration constant.

Class P (Single-particle spatial):: Electron density $\rho(r)$, radial moments $\langle r^k\rangle$, momentum distributions. Require the $S^3 \to \mathbb{R}^3$ conformal projection, introducing π through $\mathrm{Vol}(S^3) = 2\pi^2$.

Class C (Multi-particle correlated):: Correlation energies, adiabatic potential energy surfaces, multi-electron expectation values. Require embedding and flow constants ($e^a E_1(a)$, $\mu(R)$, PK pseudopotential).

VI. THE HOPF FIBRATION AND ELECTROMAGNETIC COUPLING

The results of Secs. **??**–**??** establish S^3 as the proven state manifold for a single bound electron. The three-sphere possesses a unique nontrivial fiber bundle—the Hopf fibration [**?**]—which we now examine for physical content.

The results of this section are conjectural. The numerical agreement is statistically significant but the combination rule (Eq. **??***) has not been derived from first principles. We present the evidence and identify explicitly which links in the derivation chain are established, partially established, and open.*

A. The Hopf bundle as electron-photon geometry

The Hopf fibration

$$S^1 \longrightarrow S^3 \longrightarrow S^2 \tag{17}$$

is the unique principal $U(1)$-bundle over S^2 with first Chern number $c_1 = 1$. Its three components admit a physical interpretation:

- S^3 (total space): The electron bound-state manifold, proven by Fock's stereographic projection (Sec. **??**).
- S^2 (base): The photon momentum sphere, parametrizing propagation directions of the electromagnetic field.
- S^1 (fiber): The $U(1)$ gauge phase of the electromagnetic potential.

Electromagnetic coupling is then the cost of projecting the electron's S^3 onto the photon's S^2—a projection whose geometry is entirely determined by the Hopf map.

B. Spectral invariants

Each bundle component contributes a spectral invariant.

Base S^2—Casimir trace. The SO(3) Casimir eigenvalue for angular momentum l is $l(l+1)$, with degeneracy $2l+1$. Summing over the hydrogen shells up to a cutoff $n_{\max}$:

$$B(n_{\max}) = \sum_{n=1}^{n_{\max}} \sum_{l=0}^{n-1} (2l+1)\, l(l+1). \tag{18}$$

At $n_{\max} = 3$: $B(3) = 0 + 0 + 6 + 0 + 6 + 30 = 42$. The closed-form expression is $B(m) = m(m+1)(2m+1)(m+2)(m-1)/20$.

Fiber S^1—spectral zeta function. The S^1 zeta function at $s = 1$ gives

$$F = \zeta_R(2) = \frac{\pi^2}{6} = 1.644934\ldots \tag{19}$$

Total S^3—boundary correction. At cutoff $n_{\max} = 3$, the highest eigenvalue is $|\lambda_3| = 8$ and the cumulative state count through shell 2 is $N(2) = 5$:

$$\Delta = \frac{1}{|\lambda_3| \times N(2)} = \frac{1}{40} = 0.025. \tag{20}$$

C. The selection principle

The cutoff $n_{\max} = 3$ is not arbitrary. The ratio B/N, where $N(n_{\max}) = \sum_{n=1}^{n_{\max}} n^2$ is the total state count, equals the dimension of S^3 at exactly one value:

$$\frac{B(n_{\max})}{N(n_{\max})} = \dim(S^3) = 3 \quad \Leftrightarrow \quad n_{\max} = 3. \tag{21}$$

This follows from the closed-form ratio $B(m)/N(m) = 3(m+2)(m-1)/10$; setting this equal to 3 gives $(m+4)(m-3) = 0$, with unique positive root $m = 3$. The proof is algebraic and specific to S^3: the $SO(4) \cong (SU(2) \times SU(2))/\mathbb{Z}_2$ product structure yields the clean Clebsch–Gordan branching $l = 0, \ldots, n-1$ within each shell, and $S^3 \cong SU(2)$ is the only sphere that is a Lie group.

D. The combination rule and cubic equation

The conjectural combination rule is

$$K = \pi(B + F - \Delta) = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right) = 137.036064. \tag{22}$$

The self-consistency condition $\alpha^{-1} + \alpha^2 = K$ yields the depressed cubic

$$\alpha^3 - K\alpha + 1 = 0, \tag{23}$$

whose discriminant $4K^3 - 27 > 0$ confirms three real roots. Selecting the smallest positive root (corresponding to perturbative QED):

$$\alpha^{-1} = 137.035987. \tag{24}$$

The CODATA 2018 experimental value is $\alpha^{-1} = 137.035999\ldots$, giving a relative error of 8.8×10^{-8} with zero free parameters.

E. Circulant structure

The cubic (**??**) is the characteristic polynomial of a traceless $\mathbb{Z}_3$-symmetric circulant matrix:

$$M = \begin{pmatrix} 0 & b & c \\ c & 0 & b \\ b & c & 0 \end{pmatrix}, \tag{25}$$

with $bc = K/3$ and $b^3 + c^3 = -1$. The $U(1)$ gauge structure of the Hopf fiber forces Hermiticity: $c = b^*$, so $b = \sqrt{K/3}\, e^{i\theta}$ with phase determined by $\cos(3\theta) = -1/[2(K/3)^{3/2}] \approx -0.00162$. The physical root corresponds to $\theta \approx 3\pi/2$ (nearly pure imaginary coupling), encoding a democratic interaction among three bundle components.

F. Statistical significance

To assess whether the numerical agreement is coincidental, a combinatorial search was conducted over 1.92×10^9 formulas of the form $\alpha^{-1} = T(A + B + C)$ (or cubic variants), where A, B, C are drawn from 743 spectral terms and T from 14 transcendental prefactors [**?**]. Only 10 formulas achieve precision $\leq 8.8 \times 10^{-8}$, giving

$$p\text{-value} = 5.2 \times 10^{-9}. \tag{26}$$

TABLE II. Derivation chain for α from the Hopf bundle.

Link	Description	Status
1	Hopf bundle as geometric arena	✓
2	Spectral data $\to B$, F, Δ	•
3	Combination rule $K = \pi(B + F - \Delta)$	×
4	$\mathbb{Z}_3$ circulant $\to$ cubic	∘
5	α from cubic root	✓

✓ = established; • = largely established (selection principle $B/N = 3$ proven; gap: why these specific spectral quantities); ∘ = partially established (Hermiticity proven; gap: why $bc = K/3$); × = not established (critical gap: no principle selects this additive combination).

This distinguishes the result from numerology but does not constitute a derivation.

G. Derivation chain

Table **??** summarizes the status of each link.

Link 3—the combination rule—is the critical open problem. No known spectral-geometric invariant (spectral determinant, Ray–Singer analytic torsion, Seeley–DeWitt heat kernel, Atiyah–Patodi–Singer η-invariant) reproduces K. The rule mixes a discrete combinatorial quantity ($B = 42$) with a continuous spectral invariant ($F = \pi^2/6$) in a manner without precedent in the spectral geometry literature.

Negative result: Applying the combination rule to the three higher-dimensional Hopf fibrations ($S^3 \to S^7 \to S^4$, $S^7 \to S^{15} \to S^8$) does not yield known coupling constants. Only the complex Hopf fibration produces a physically meaningful result [**?**].

VII. ATOMIC SPECTRA: A TESTABLE PREDICTION PROGRAM

A. Non-relativistic energies from S^{3N-1}

For single-center systems, the hyperspherical framework on S^{3N-1} is mathematically exact and computationally convergent. Current results include He at 0.05% (single adiabatic channel on S^5, Paper 13 [**?**]), Li at 1.10%, and Be at 0.90% (multi-electron FCI).

The coupled-channel framework, already prototyped through $l_{\max} = 5$ with algebraic convergence $\sim l_{\max}^{-2}$ to a structural floor of 0.19–0.20% [**?**], provides a clear path to higher precision. The literature benchmark for hyperspherical He is 0.1 ppm with 11 adiabatic channels [**?**].

B. Fine structure as a test of the α conjecture

The non-relativistic energy E_{NR} computed from S^{3N-1} can be compared with the experimental energy:

$$E_{\mathrm{exp}} = E_{\mathrm{NR}} + E_{\mathrm{rel}}(\alpha) + E_{\mathrm{QED}}(\alpha) + \cdots \tag{27}$$

If E_{NR} is computed to sufficient precision, the residual $E_{\mathrm{exp}} - E_{\mathrm{NR}}$ isolates the α-dependent relativistic and QED corrections. Fine structure splitting scales as α^2; the Lamb shift as α^5. An independent determination of α from geometric atomic spectra would test the Hopf conjecture computationally.

C. Milestones

A concrete research program proceeds in four phases:

1. **He ground state to <1 ppm.** Coupled channels on S^5 with 5–15 channels. Existing infrastructure: algebraic spectral basis, Hellmann–Feynman P-matrix, exact Q-matrix. Bottleneck: extending beyond 5 channels; may require non-adiabatic coupling.

2. **He excited states and transition energies.** Separate adiabatic potential curves for ^1S and ^3S states. Transition energies are more precisely measured than absolute energies; residuals vs. experiment constrain α^2 corrections.

3. **α extraction.** Compute non-relativistic He energy to precision better than the fine structure splitting ($\sim$0.36 cm^{-1} for He $2\,^3P$). Test: does the extracted α match 1/137.035987 (Hopf) vs. 1/137.035999 (CODATA)? The 8.8×10^{-8} difference corresponds to $\sim$0.0001 cm^{-1} in He fine structure.

4. **Li and Be.** Ground-state energies to $< 0.01\%$ via coupled channels on S^8 (Li, 3 electrons) and S^{11} (Be, 4 electrons). The H-set coupling from Sec. **??** provides the angular framework for Be. Multi-electron convergence validates the S^{3N-1} framework beyond He.

VIII. SCOPE AND HONEST LIMITATIONS

A. What the framework does

1. Derives quantum number structure $(l, m, 2l + 1)$ from geometric axioms, independent of the Schrödinger equation.

2. Provides a mathematically exact formulation for single-center Coulomb systems, verified through 18 symbolic proofs.

3. Produces computationally efficient qubit encodings: $O(Q^{2.5})$ Pauli scaling for composed molecular Hamiltonians, 51–1,712× fewer Pauli terms than Gaussian baselines [**?**].

4. Identifies a new sparsity mechanism ($6j$ recoupling) that is basis-agnostic.

5. Classifies transcendental content through the exchange constant taxonomy.

6. Suggests a topological origin for α with zero free parameters and $p = 5.2 \times 10^{-9}$ (conjectural).

B. What the framework does not do

1. Replace computational quantum chemistry for molecules. The molecular limitation is fundamental: the multi-center N-body problem does not separate in any known coordinate system.

2. Provide convergent molecular potential energy surfaces. The composed approximation has characterized accuracy ceilings (5.3–26% equilibrium distance error).

3. Derive the combination rule for α (Link 3 remains open).

4. Account for relativistic effects. The framework is non-relativistic; fine structure enters as a correction (Sec. **??**).

5. Explain why S^3 is the physical manifold. The framework takes Fock's 1935 result as given and builds upon it.

C. The molecular boundary

Six nested approaches to molecular electronic structure within the S^{3N-1} framework have been tested: single-center, charge-center, heterogeneous Löwdin, and three variants of the composed architecture. All single-basis approaches fail for heteronuclear molecular potential energy surfaces. The root cause is that core and valence electrons occupy different length scales; any unified-basis approach faces orthogonalization penalties that destroy the sparsity advantage. The composed architecture [**?**]—which assigns each electron group its own natural coordinate system—provides practical molecular Hamiltonians with $N_{\mathrm{Pauli}} \approx 11.11 \times Q$ across 30 molecules.

This molecular limitation does not affect the single-center mathematical structure, just as turbulence does not invalidate the Navier–Stokes equations.

IX. CONCLUSION

The kinematic structure of non-relativistic quantum mechanics—quantum numbers, degeneracies, selection rules—emerges from the spectral theory of Laplace–Beltrami operators on compact manifolds, specifically the unit hyperspheres S^{3N-1} for N-electron systems. The dynamic structure—energy levels, the Coulomb potential—enters through projection parameters that map the dimensionless topology into physical observables. Transcendental content is classified by the exchange constant taxonomy, appearing at precisely identifiable steps in the projection chain.

For single-center systems, these results are mathematically proven and computationally validated: 18 symbolic proofs for S^3, 0.05% He ground state on S^5, chemical periodicity from S_N representation theory on S^{3N-1}, and a new $6j$ recoupling sparsity mechanism reducing ERI density by 18–26%.

The Hopf fibration of the electron's S^3 state manifold suggests—but does not yet prove—that the electromagnetic coupling constant α is determined by the same topology. A systematic program of high-precision atomic spectra computation on S^{3N-1}, compared with experimental fine structure data, offers a concrete and testable path forward.

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